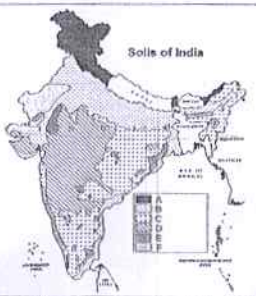
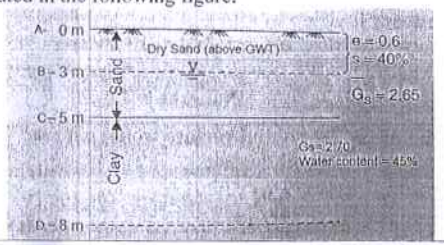
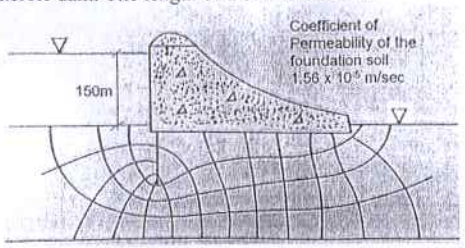
Roll No. | | | | | | | |

EVEN END SEMESTER EXAMINATION JUNE – 2025

Name of the Course: B. Tech (Civil Engineering)
Semester: IV
Name of the Paper: SOIL MECHANICS
Paper Code: TCE403
Time: 3 Hours Maximum Marks: 100

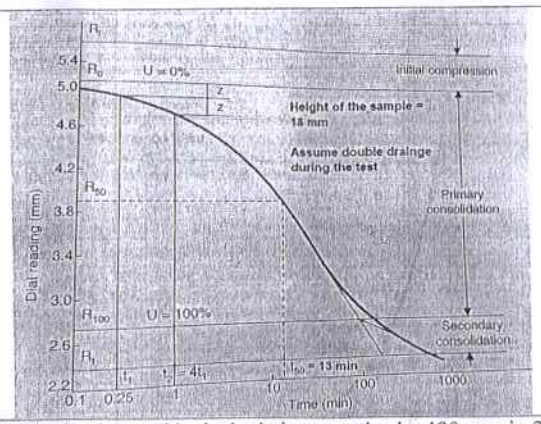
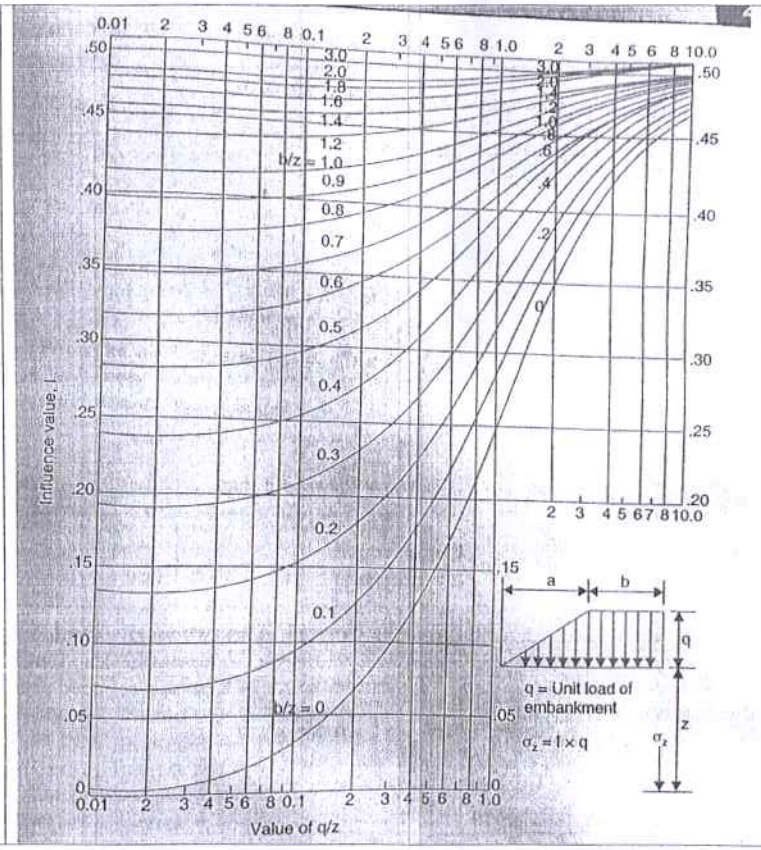
Note:-
(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty and each question carries 10 marks

Q1	**(20marks)**	
(a)	 Identify A, B, C, D, E and F soil type from the regional soil deposit map of India as provided. Discuss in detail the specific characteristics of the Black cotton soil deposit found in Indian subcontinent.	CO1
(b)	Explain the following transported soil based in the transporting agent and method of deposition • Alluvial Deposit • Lacustrine Deposit • Marine Deposit • Aeolian Deposit • Glacial Deposit	
(c)	A saturated clay sample originally 25 mm thick and at a initial void ratio of 1.12 was subjected to a compressive load. After the clay sample was completely consolidated, its thickness was measured to be 23.5 mm. Compute the final void ratio of the soil sample.	
Q2	**(20 marks)**	
(a)	Calculate Total stress, Effective stress and pore water pressure values at points A, B, C, and D as indicated in the following figure. 	CO2
(b)	For a field pumping test, a well was sunk through a horizontal stratum of sand 14.5 m thick and underlain by a clay stratum. Two observation wells were sunk at horizontal distances of 16 m and 34 m respectively from the pumping well. The initial position of the water table was 2.2 m below ground level. At a steady state pumping rate of 925 litres/minute, the drawdowns in the observation wells were found to be 2.45 m and 1.20m respetively. Calculate the coeffcient of permeability of the sand.	
(c)	With reference to the following fgure, estimate the seepage discharge through the foundation of concrete dam. The length of the dam is 250m. 	
Q3	**(20 marks)**	
(a)	The *in situ* void ratio of a granular soil deposit is 0.50. The maximum and minimum void ratios of the soil were determined to be 0.75 and 0.35. Specific gravity of the soil solid is 2.67. Determine the relative density and relative compaction of the soil deposit.	CO3
(b)	The soil from a borrow area having an average in situ unit weight of 15.5 kN/m3 and water content of 10% was used for the construction of an embankment with total finished volume 60,000 cum. In half of the embankment, due to improper control during rolling, the dry unit weight was slightly lower. If the dry unit weight achieved in the two parts is 16.5 kN/m³ and 21.5 kN/m³, calculate the amount of saving done by the contractor. Cost of soil excavation from the borrow area including its transportation to the site is 1500 per cum.	
(c)	The following data refers to a compaction test as per Indian Standard (light compaction):	

Water Content (%)	8.5	12.2	13.75	15.5	18.2	20.2
Weight of wet sample (kg)	1.80	1.94	2.00	2.05	2.03	1.98

If the specific gravity of the soil grains was 2.70, obtain the MDD and OMC values from the compaction curve. Also, if RC to be achieved in the field is 90%, estimate the range of moisture content to be allowed in the field compaction.

Q4	**(20 marks)**	
(a)	Calculate the coefficient of consolidation (in cm²/s) of the soil from the graph provided pertaining to Casagrande Method	CO4



(b)	A 8 meter thick clay layer with single drainage settles by 120 mm in 2 years. The coefficient of consolidation for this clay was found to be 6×10-3 cm²/sec. Calculate the likely ultimate consolidation settlement and find out how long it will take to undergo 90% of this settlement.	
(c)	A normally consolidated clay layer settled by 20 mm when the effective stress was increated from 25 kPa to 50 kPa. What will be the settlement when the effective stress is increased from 50 kPa to 100 kPa.	
Q5	**(20 marks)**	**CO5**
(a)	In a CU test on a normally consolidated clay, a sample consolidated under a stress of 200 kPa failed at an additional axial stress of 150 kPa. The pore pressure at failure was 75 kPa. Determine analytically the shear strength parameters both in terms of total and effective stress.	
(b)	In a direct shear test conducted on a dense sand, the sample fails at a shear stress of 75 kPa, when the normal stress was held constant at 100 kPa. Determine the angle of shear resistance, orientation of major and minor principal planes as well as orientation of plane of maximum shear stress. Use Mohr-circle diagram for the same.	
(c)	Using the following diagram, estimate the amount of stress developed at depth 10 m due to embankment loading. Take a = 5m and b = 3m, q = 250 kPa.